"Food Insecurity In Africa - Examples From Bangladesh and India"

Ques: Why food insecurity persists in Africa? Give an example in the context of Bangladesh's Shrimp Industry.

Answer: Pre-mercantilist Africa was self-sufficient in food production and was neither inferior nor weaker than the rest of the Ancient World. Food production was the primary goal of agriculture. Sophisticated techniques had evolved which were suited to the needs of the people and their fragile environment and which helped them cope with cyclical droughts and crop failures due to other causes. Adequate food security and self-reliance were common before European colonization. Methods had been developed for storage of surpluses and complex patterns of trade had been developed around food availability.

The British colonial agricultural system affected the African traditional agricultural system and and led the African to food insecurity persistent.

Colonial Transformation: Growing demand from industries in Europe and North America drove exponential growth in export crops at the expense of food security. In Western Africa, exports to Britain and France doubled five times and three times, respectively, over a 70-year period (1870 to 1940). The area dedicated to export crops in SSA increased by 56% during the 1800s and a further 136% from 1900 to 1960. This increase was initially fuelled by high commodity prices, but after 1880 it was driven by the need to of-set the impact of declining prices and the onset of WWI. Hence, the growth in export crop production was driven by colonial demand rather than farmers responding to market opportunities. That is, the expansion occurred due to the colonial demand, not for the farmers responding to manket opportunities.

Peasants preffered to grow edible crops (wheat, nice, potato etc.) for consumption. But forced to grow non-edible crops (tea, coffee, cotton) for export. At first, not only for foreign needs, some of the farmers were also willing to produce cash crop for the bigger super store or depending on the cash incentives in the market situation. But later, many were forced to produce cash crop over food crop which led to the reduction in food production. Farmers had to clear marginal, unfertile kind of land to use it to produce household food.

The expansion in the growth of cash crop increased labor demand in cash crop but the availability of manpower is reduced to produce staple foods. So, food-supply diminished as well as livestock production had also marginalized.

Shrimp Industry: An Example From Bangladesh.

(Which Altered The Traditional Land, Lobr and Agricultural System) ____ For nearly 50 years, shrimp farming has brought short-term profits to Bangladesh. But now, with climate change, it's harming the environment and local communities. By 2050, sea levels could rise by 50 cm, threatening the crowded coastal areas where people keep moving, even as risks grow.

Land: In Bangladesh, rising sea levels could cause major land loss and destroy livelihoods as saltwater spreads into rivers and water sources that once supported farming, animals, and people. Climate change makes this worse, and years of shrimp farming have sped up the spread of saltwater.

From 1973 to 2012, Bangladesh's shrimp exports grew from $2.9 million to $590 million. The industry took off after independence in 1971, attracting many businesses. But now, earnings are falling as shrimp production drops due to disease, which is common in intensive farming.

As shrimp farming has expanded, it has made the soil and water in southwestern Bangladesh saltier, while profits and harvests have declined. To farm shrimp, land that once grew rice was changed. Salty water was brought in, sometimes by breaking embankments that used to protect rice fields, ponds, and homes.

Shrimp Farming; Change in Traditional Agricultural System: According to the Bangladesh Frozen Foods Exporters Association (BFFEA), today shrimp are being cultivated in 276,000 hectares of the 337,164 hectares of land available along the coast, or 81%. Of them, 195,000 hectares or 70.6% are in Satkhira, Khulna and Bagerhat districts.

Labor: As shrimp farms surrounded small-holdings, the saline water leached into their land. This killed the rice plants in the paddies, even when the embankments were not deliberately damaged. Gradually, most farmers moved from rice to shrimp cultivation, many in the hope of making more money, some because they had no choice.

The process was like that - Influential people started the cultivation, then a growing number of small-holders, previously happy with cultivating paddy, were drawn to shrimp farming in the hope of increasing their profits. Others were forced to switch as their land became saline due to surrounding shrimp farms.

Water Crisis/Environmental Impact: Where there is shrimp culture, the freshwater crisis is more acute”. Ponds are the major source of freshwater in rural Bangladesh. When farms nearby turn saline due to shrimp aquaculture, brackish water seeps into the ground and contaminates the ponds, which become saline as well.

A 2019 study by WaterAid Bangladesh found that many ponds in Shyamnagar, Satkhira, had water too salty for drinking or cooking. Of 57 ponds, 16 were too salty for any use, 25 could be used only for cooking, and 14 were reserved for shrimp farming. Similar issues were reported in Khulna and Bagerhat districts.

With freshwater becoming scarcer, coastal residents collect rainwater during the June–September monsoon and stretch it as long as they can. After that, they have to buy drinking water at 40 times the price people in Dhaka pay.

Many people in poor households drink very little water during the dry months, which harms their health.

Parallel Incident From The Colonial History Of Africa Toward India.

Since India adopted economic reforms in 1991, problems related to food production and access have worsened. In recent years, slow agricultural growth and rising unemployment have made the situation more serious. Grain production and consumption per person have dropped, while income has increasingly shifted from the majority of people to a small wealthy group. As a result, many people are eating less and can't afford new clothes.

To understand food problems, it's helpful to look at both long-term and short-term causes. These issues aren't just in India but also affect other developing countries like China. One major issue is using food grains to feed animals instead of people. This reduces the food available to the poor and is made worse by growing inequality—both within countries and between rich and poor nations. Free trade also pressures poor countries to grow export crops instead of food for local people.

In medieval Europe, low farm productivity meant there wasn't enough grain for both people and animals, especially in winter. This led to many animals being slaughtered seasonally and people eating lots of salted meat. Even after the agricultural revolution in the 18th and 19th centuries, Europe still struggled with low productivity. To meet their needs, Europeans increasingly imported wheat and other goods like sugar, rice, tea, and cotton from colonies. These colonies, mostly in tropical regions, had to shift land away from growing food for their own people to produce these exports, leading to less food for local populations.

Today's competition between food and animal feed is different but still follows a similar pattern.

Because of Industrialization and urbanization, used in Hotel and restaurant (fast food and meat menu), rise of middle class/upper middle class, Increased in female labor participation in workforce (impacts the cooking habits which increased the demand of processed and preservative item like meat), the global meat demand has increased.

In India, meat production—especially of beef and buffalo meat—has seen significant changes over the decades, and European consumption has played a role in influencing land-use patterns. Here's an explanation:

1. Increased Export Demand: European countries have imported large quantities of buffalo meat from India. To meet this demand, India expanded its meat production sector, particularly in regions like Uttar Pradesh, Maharashtra, and Andhra Pradesh. According to recent data, India is the world's fifth largest meat exporter. The growing demand from countries in the Middle East, Southeast Asia, and parts of Europe has incentivized large-scale buffalo rearing and meat processing. This shift prioritizes meat production over traditional mixed agriculture or subsistence farming.

2. Shift in Land Use: To support meat production, more land has been allocated to grow fodder crops instead of traditional food crops.

3. Grazing Land Issue: Before massive Industrialization, there were no need to produce grass. Grass was available. So, the production cost of grass was zero. Massive idustrialization left less of grazing land. Thereby, Grazing land has been expanded or repurposed, often at the expense of forests or common lands. As a result of less grazing land, natural grass is replaced by stall feeding. Production cost increased and this led to the establishment of another industry - Feed Industry.

4. Feed Issue: Poorer tropical countries are used to grow feed and animal products for wealthy nations. Grains that could feed people are instead used to raise animals, which is wasteful.

For example, 1 kg of beef gives 1,140 calories and 226 grams of protein, but the grain used to produce that beef could give 24,150 calories and 700 grams of protein if eaten directly.

Another study shows -

• For 1 kg of extra meat, 7 kg of food grain is needed.

• For 1 kg of poultry meat, 2 kg of food grain is needed.

• For 1 litre of extra milk, 0.2 kg of food grain is needed.

Feed production has significantly increased in India, with a production at a rising rate of seven to eight times. India is the second-largest animal feed producer in 2023. The production again came from the expense of food production. Food crop land was replaced by feed production.

5. Impact on Small Farmers: Land-use changes often marginalize small-holders, as large companies or exporters dominate meat production and control larger land areas.

6. Economic and Environmental Consequences: Both in the cases of African volonization and Indian meat production show how global market integration, driven by European (or broader Western) consumption, leads to:

• Monoculture or mono-use of land.

• Decreased biodiversity.

• Displacement of small-scale or indigenous practices.

• Increased export orientation over food self-sufficiency.

• Greater environmental stress, including land degradation and water pollution from concentrated animal waste.

In short, the growing European appetite for Indian meat—mainly buffalo—has transformed parts of India’s agricultural landscape, shifting land use from subsistence crops to fodder and meat processing infrastructure. This reflects the global link between consumption patterns in developed nations and land-use changes in developing countries.

Ques: If the government of Bangladesh officially started to encourage meat export, as a political economist, how you see the decision. Critically evaluate the decision.

Answer: Encouraging meat exports from Bangladesh raises complex political and economic considerations. While potentially boosting foreign exchange and economic growth, it also risks undermining domestic food security, potentially harming the environment, and could lead to inconsistencies with Bangladesh's existing agricultural policies. Here's a critical evaluation is given _____

1. Economic Growth and Trade Balance.

Benefit: Bangladesh's meat export could contribute significantly to GDP growth. By tapping into international meat markets, Bangladesh can increase its export revenues, helping to reduce trade deficits.

Drawback: However, Bangladesh would need to ensure that it has the infrastructure and quality controls in place to meet international standards, especially in high-demand markets such as the EU, Middle East, or Southeast Asia. This requires large investments in cold chains, transportation, and processing facilities.

2. Domestic Supply and Agricultural Impact.

Benefit: Encouraging meat exports could also benefit the domestic agriculture sector by increasing the demand for livestock, which would provide higher prices for farmers, especially those in rural areas. This could lead to increased incomes for farmers, stimulate rural economies, and provide more incentives for sustainable livestock farming.

Drawback: However, an emphasis on meat exports could lead to domestic supply shortages, resulting in higher prices for meat within the country.

Data: According to a recent study (2022), in Bangladesh, the annual meat consumption per capita is only 4.63 kg. If the export market becomes more competitive, domestic consumers could face higher prices, hurting their purchasing power & the consumption level can further decrease.

3. Job Creation.

Benefit: A burgeoning meat export sector could create jobs in areas like livestock farming, processing, and logistics.

Drawback: Shifting labor toward export production can lead to the less of available manpower in food production.

Moreover, the higher initial cost can lead to labor exploitation (low wage, poor working condition) like the garment industry.

4. Environmental Considerations.

Benefit: From an environmental standpoint, Bangladesh could emphasize sustainable and ethical farming practices that improve the global image of its meat exports, positioning them as premium products.

Drawback: The environmental impact of increasing meat production is significant. Meat production is resource-intensive, requiring large amounts of land, water, and feed. This could worsen environmental degradation, including deforestation, soil erosion, and water scarcity, which are already significant challenges for Bangladesh.

The FAO has warned that the livestock sector contributes to about 14.5% of global greenhouse gas emissions. If Bangladesh expands its meat exports without adopting sustainable practices, it could inadvertently worsen these environmental challenges.

5. Infrastructure and Investment Challenges.

Benefit: To develop a competitive meat export sector, Bangladesh would need to invest heavily in the meat-processing and cold storage infrastructure, which could generate employment opportunities and improve technological development in rural areas.

Drawback: On the flip side, the initial costs of building infrastructure are enormous. Bangladesh’s existing infrastructure is not yet sufficient to support large-scale meat exports.

6. Policy Inconsistencies.

Prioritizing meat exports might clash with policies aimed at promoting food production for domestic consumption and sustainable agriculture.

7. Grazing Land & Feed Issue. [Same explanation as India's case. Feed company: Kazi Farm, CP, Provita etc.]

Encouraging meat exports in Bangladesh presents both opportunities and challenges. On the one hand, it could diversify the economy, boost rural incomes, and promote industrial growth. On the other hand, it could lead to higher domestic meat prices, environmental degradation, public health risks, and labor exploitation without careful regulation.

To make this decision sustainable, the government would need to:

1. Develop infrastructure.
2. Implement sustainable practices.
3. Ensure regulatory frameworks.
4. Balance domestic and export markets.